



10.0.1 ISRO-2013-76



The number of edges in a 'n' vertex complete graph is?

- A. $n^*(n-1)/2$ B. n^2
C. $n^*(n+1)/2$ D. $n^*(n+1)$

isro2013 graph-theory

Answer key 

10.0.2 ISRO CSE 2023 | Question: 63



Maximum number of Simple graphs possible with n vertices

- A. $2^{n(n-1)/2}$ B. $2^{(n-1)/2}$ C. $2^{(n(n+1)/2)}$ D. $2^{n(n+1)}$

isro-cse-2023 graph-theory discrete-mathematics

Answer key

10.0.3 ISRO CSE 2023 | Question: 51



Which of the following are true?

- i. In a graph G with 'n' vertices and 'e' edges, sum of degrees of vertices = $2 \cdot e$.
 - ii. Eccentricity of a connected graph can never be equal to radius of the graph
 - iii. Girth of a graph is the shortest cycle of the graph
 - iv. Graph with equal degree for all vertices is multigraph
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- a. (i), (ii), (iii)
 - b. (ii), (iii), (iv)
 - c. (i), (iii), (iv)
 - d. None of the above

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Answer key

10.1 Degree of Graph (1)

10.1.1 Degree of Graph: ISRO CSE 2009 | Question: 52



If G is a graph with e edges and n vertices the sum of the degrees of all vertices in G is

- A. e B. $e/2$ C. e^2 D. $2e$

isro2009 graph-theory degree-of-graph

Answer key

10.2 Euler Graph (2)

10.2.1 Euler Graph: ISRO CSE 2007 | Question: 06



A graph with n vertices and $n - 1$ edges that is not a tree, is

- A. Connected B. Disconnected C. Euler D. A circuit

isro2007 graph-theory euler-graph

Answer key

10.2.2 Euler Graph: ISRO CSE 2016 | Question: 5



A given connected graph G is a Euler Graph if and only if all vertices of G are of

- A. same degree
B. even degree
C. odd degree
D. different degree

graph-theory euler-graph isro2016

Answer key

10.3 Graph Coloring (1)

10.3.1 Graph Coloring: ISRO CSE 2007 | Question: 07



If a graph requires k different colours for its proper colouring, then the chromatic number of the graph is

- A. 1 B. k C. $k - 1$ D. $k/2$

isro2007 graph-theory graph-coloring

Answer key

10.4 Graph Connectivity (10)

10.4.1 Graph Connectivity: GATE CSE 1994 | Question: 1.6, ISRO2008-29



The number of distinct simple graphs with up to three nodes is

- A. 15 B. 10 C. 7 D. 9

gate1994 graph-theory graph-connectivity combinatory normal isro2008 counting

Answer key

10.4.2 Graph Connectivity: GATE CSE 2002 | Question: 1.25, ISRO2008-30, ISRO2016-6



The maximum number of edges in a n -node undirected graph without self loops is

- A. n^2 B. $\frac{n(n-1)}{2}$ C. $n - 1$ D. $\frac{(n+1)(n)}{2}$

gatecse-2002 graph-theory easy isro2008 isro2016 graph-connectivity

Answer key

10.4.3 Graph Connectivity: GATE CSE 2003 | Question: 8, ISRO2009-53



Let G be an arbitrary graph with n nodes and k components. If a vertex is removed from G , the number of components in the resultant graph must necessarily lie down between

- A. k and n B. $k - 1$ and $k + 1$
C. $k - 1$ and $n - 1$ D. $k + 1$ and $n - k$

gatecse-2003 graph-theory graph-connectivity normal isro2009

Answer key

10.4.4 Graph Connectivity: ISRO CSE 2007 | Question: 62



Let X be the adjacency matrix of a graph G with no self loops. The entries along the principal diagonal of X are

- A. all zeros
- B. all ones
- C. both zeros and ones
- D. different

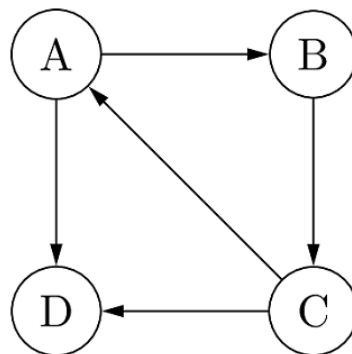
isro2007 graph-theory graph-connectivity

Answer key

10.4.5 Graph Connectivity: ISRO CSE 2008 | Question: 78



Consider the graph shown in the figure below:



Which of the following is a valid strong component?

- A. a, c, d
- B. a, b, d
- C. b, c, d
- D. a, b, c

isro2008 graph-theory graph-connectivity

Answer key

10.4.6 Graph Connectivity: ISRO CSE 2009 | Question: 54



A graph in which all nodes are of equal degree, is known as

- A. Multigraph
- B. Non regular graph
- C. Regular graph
- D. Complete graph

isro2009 graph-theory graph-connectivity

Answer key

10.4.7 Graph Connectivity: ISRO CSE 2009 | Question: 55



In a graph G there is one and only one path between every pair of vertices then G is a

- A. Path
- B. Walk
- C. Tree
- D. Circuit

isro2009 graph-theory graph-connectivity

Answer key

10.4.8 Graph Connectivity: ISRO CSE 2009 | Question: 56



A simple graph (a graph without parallel edge or loops) with n vertices and k components can have at most

- A. n edges
- B. $n - k$ edges

C. $(n - k)(n - k + 1)$ edges

isro2009 graph-theory graph-connectivity

Answer key

D. $(n - k)(n - k + 1)/2$ edges



10.4.9 Graph Connectivity: ISRO CSE 2011 | Question: 35

How many edges are there in a forest with v vertices and k components?

A. $(v + 1) - k$

B. $(v + 1)/2 - k$

C. $v - k$

D. $v + k$

isro2011 graph-theory graph-connectivity

Answer key



10.4.10 Graph Connectivity: ISRO CSE 2018 | Question: 38

The number of edges in a regular graph of degree: d and n vertices is:

A. maximum of n and d

B. $n + d$

C. nd

D. $nd/2$

isro2018 graph-theory graph-connectivity

Answer key

10.5

Qod Cse 23 (1)

10.5.1 Qod Cse 23: ISRO CSE 2023 | Question: 52

If there are five faces and nine vertices in an undirected planar graph, then number of edges is

A. 14

B. 16

C. 12

D. None of the above

isro-cse-2023 graph-theory graph-planarity qod-cse qod-cse-23

Answer key



Answer Keys

10.0.1	A	10.0.2	A	10.0.3	D	10.1.1	D	10.2.1	B
10.2.2	B	10.3.1	B	10.4.1	C	10.4.2	B	10.4.3	C
10.4.4	A	10.4.5	D	10.4.6	C	10.4.7	C	10.4.8	D
10.4.9	C	10.4.10	D	10.5.1	C				